Physics-Aware Hybrid MCMC and Swin-Transformer Framework for Grouped Inversion of Groundwater and Shallow Hydrologic Storage from InSAR Deformation

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***Abstract*—A novel physics-aware hybrid MCMC and Swin-transformer framework is presented for grouped inversion of groundwater and shallow hydrologic storage from InSAR land-deformation time series. The method is designed to perform layered hydrologic inference directly from mixed high-resolution deformation data, without any prior explicit filtering of disturbing surface or subsurface processes and without area-specific categorization.**

The approach is first proven in a controlled synthetic five-layer setting, where the clean Stage 1 benchmark achieves approximately $R^2 = 0.976$ for load-related storage, $R^2 = 0.980$ for total water storage, $R^2 = 0.980$ for deformation reconstruction, and $R^2 = 0.981$ for groundwater. For real data, conditioning analysis supports a grouped inversion in Emilia-Romagna, Italy, reducing the stacked-operator condition number from approximately 2.16×10^6 to 6.99×10^1 and restoring practical rank from $4/5$ to $3/3$. In the applied Bologna case, the grouped multisensor posterior $[S0 + Ss,\ Sd + Sr,\ Sg]$ achieves regional posterior skill of approximately $R^2 = 0.971$ for InSAR, $R^2 = 0.487$ for GRACE, and $R^2 = 0.618$ for SMAP, with tiled InSAR posterior skill of approximately $R^2 = 0.905$.

Independent validation against groundwater wells supports the groundwater counterpart of the grouped inversion, with 83 evaluated station series, a median best-match anomaly correlation of about 0.704, and a conservative trusted subset of 21 stations across 6 hydrogeologic groups. The shallow hydrologic component is recovered at higher spatial resolution and is supported through physically plausible posterior structure and multisensor consistency. These results support a practical route toward automated downstream satellite retrieval of regional and local hydrologic storage.

***Index Terms*—InSAR deformation, grouped inversion, groundwater storage, shallow hydrologic storage, hydrologic storage inversion, Bayesian state-space modeling, Markov chain Monte Carlo, physics-aware transformers, Swin3D U-Net.**

I. Introduction

GROUNDWATER monitoring increasingly requires retrieval systems that work at aquifer-relevant scale rather than only at the coarse support of basin-scale storage products. GRACE and GRACE-FO have transformed terrestrial water-storage observation, and Bayesian assimilation frameworks such as MCMC-DA have shown how large-scale storage observations can be fused with hydrologic models [1], [2]. However, these products remain too coarse for direct aquifer-scale diagnosis in heterogeneous and intensively managed settings, while direct subsurface observations remain sparse [3]–[5].

InSAR is attractive in this context because it resolves millimeter-scale land deformation at high spatial resolution, making it one of the few satellite observables that can sense local hydrologic loading, aquifer compaction, and groundwater-related deformation at regional to local scale [6]–[9]. Emilia-Romagna is characterized by groundwater variability, shallow loading, subsidence, and intensive water use, together with a highly monitored multilayer aquifer setting. This makes the region a relevant test bed for the framework and for assessing the success of the grouped layered inference, particularly for the groundwater counterpart [10]–[12]. The challenge is that the InSAR signal is small, noisy, and mixed with other geophysical contributions, so inversion strategies developed for coarse storage time series are not directly applicable to high-resolution deformation [1], [2], [7].

The novelty of this work is a grouped layered inversion framework that retrieves groundwater and shallow hydrologic storage directly from mixed InSAR deformation without prior explicit filtering of disturbing surface or subsurface processes. The method is first checked in a synthetic five-layer setting and is then applied to Emilia-Romagna using grouped multisensor inference. The strongest supported real-data result is a grouped posterior that separates shallow loading, deep loading, and groundwater and is independently supported by groundwater-well anomalies.

II. Study Area and Data

The applied study domain is the Bologna overlap within Emilia-Romagna, northern Italy, with working bounding box longitude 10.2–12.5 and latitude 43.8–45.9 over the common MintPy/W3RA period 2017-01-04 to 2024-08-01. The main observational input is the MintPy InSAR time series derived from Sentinel-1 and processed through a custom Docker-based interferometric workflow built on ISCE++ with GPU/CUDA adaptation, followed by MintPy inversion [13]. W3RA provides the hydrologic state variables $S0$, Ss, Sd, Sg, and Sr, which are linearly interpolated onto the InSAR grid and used to define the grouped latent state [1]. GRACE/GRACE-FO

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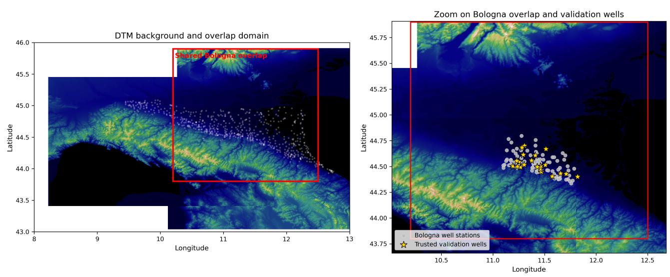

Fig. 1. Study area and validation setting. Left: DTM background over the broader Bologna–Emilia-Romagna scene with the shared InSAR/W3RA overlap marked by a red box. Right: zoom on the overlap domain showing the validation-well network and the trusted well subset used for interpretation.

JPL RL06.3 mascon anomalies provide a regional storage constraint [14], [15], SMAP provides a shallow-hydrologic constraint [16], SWOT is included as an exploratory surface-water constraint [17], [18], and ARPAE manual and automatic wells provide independent validation [11], [12].

Before assimilation, the MintPy deformation series is converted to anomaly space, W3RA is interpolated onto the InSAR grid and restricted to the shared period, GRACE and SMAP are reduced to regional anomalies, SWOT is nearest-date matched, and the wells are merged with metadata and converted to anomaly-space series. Figure 1 shows the study area and validation setting.

III. Method

The framework combines deformation-space Bayesian inversion with physics-aware Swin-based residual learning. Let t denote time and p spatial grid cell. The full latent hydrologic state is

$$\mathbf{z}_{t,p}^{\mathsf{T}} = \begin{bmatrix} S0_{t,p} & Ss_{t,p} & Sd_{t,p} & Sg_{t,p} & Sr_{t,p} \end{bmatrix}, \tag{1}$$

where the load-driven and groundwater-pressure terms are

$$\Delta\ell_t = \rho_w \left(S0_t + Ss_t + Sd_t + Sr_t \right), \qquad \Delta p_t = \rho_w g \frac{Sg_t}{S_{\mathrm{eff}}}, \tag{2}$$

and the total modeled deformation is obtained from elastic loading and poroelastic response operators,

$$u_t^{\mathrm{tot}} = G_{\mathrm{load}} * \Delta\ell_t + G_{\mathrm{poro}} * \Delta p_t. \tag{3}$$

A component-wise deformation predictor is then built as

$$\mathbf{Z}_{t,p} = \begin{bmatrix} u_{t,p}^{(S0)} & u_{t,p}^{(Ss)} & u_{t,p}^{(Sd)} & u_{t,p}^{(Sg)} & u_{t,p}^{(Sr)} \end{bmatrix}, \tag{4}$$

so that Stage 1 is posed in deformation space,

$$Y_{t,p} = \mathbf{Z}_{t,p}\boldsymbol{\theta}_{t,p} + \varepsilon_{t,p}, \qquad \mathbf{x}_{t,p}^{\mathrm{prior}} = \boldsymbol{\theta}_{t,p} \odot \mathbf{z}_{t,p}, \tag{5}$$

where $Y_{t,p}$ is the observed deformation anomaly and $\mathbf{x}_{t,p}^{\mathrm{prior}}$ is the posterior corrected prior state. The mathematical foundations, posterior existence, and forward operators are given in Supplementary Appendices A and B.

Algorithm 1 Physics-aware hybrid MCMC–Swin workflow for grouped hydrologic inversion

Require: InSAR time series, W3RA states, GRACE, refreshed SMAP, exploratory SWOT, wells
Ensure: Grouped posterior fields, multisensor fits, and validation diagnostics

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Build the common Bologna overlap and convert the InSAR series to anomaly space
Interpolate W3RA states onto the InSAR grid over the shared period
Construct deformation predictors using (2)–(6)
for each synthetic setting do
    Generate five-layer synthetic storage fields and forward deformation
    Run Stage 1 Bayesian recovery using (7)–(9)
    if Stage 2 is enabled then
        Learn residual corrections using (14)–(15)
    end if
end for
Define the grouped real-data state using (10)
for each time step t in the Bologna overlap do
    Assemble the multisensor observations
    Advance the grouped prior using (11)
    Update the grouped state using (12)
end for
if Stage 2 refinement is used then
    Apply residual learning and physics closure using (14)–(17)
end if
for each well station do
    Interpolate the grouped posterior, align anomalies, and search validation lags
end for
return grouped posterior maps, regional fits, and well-validation summaries
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For the applied Emilia-Romagna case, conditioning diagnostics support grouped inversion rather than unconstrained five-layer inversion. The grouped state is

$$\mathbf{x}_t^{\mathrm{grp}} = \begin{bmatrix} \mathrm{ShallowLoad}_t \\ \mathrm{DeepLoad}_t \\ \mathrm{Groundwater}_t \end{bmatrix} = \begin{bmatrix} S0_t + Ss_t \\ Sd_t + Sr_t \\ Sg_t \end{bmatrix}, \tag{6}$$

and the multisensor observation model is

$$\mathbf{y}_t = H_t \mathbf{x}_t^{\mathrm{grp}} + \boldsymbol{\epsilon}_t, \tag{7}$$

with InSAR, GRACE, refreshed SMAP, and exploratory SWOT entering through grouped observation operators summarized in Supplementary Appendix B. A Swin-transformer residual learner is then used as an optional Stage 2 correction,

$$\hat{\mathbf{r}}_{t,p} = f_\phi(\mathbf{I}_{t-w+1:t,\mathcal{N}(p)}, \mathbf{C}_{\mathcal{N}(p)}, \mathbf{x}_{t,\mathcal{N}(p)}^{\mathrm{prior}}, \boldsymbol{\Sigma}_{t,\mathcal{N}(p)}^{\mathrm{prior}}), \tag{8}$$

followed by physics closure through the same hydromechanical operator. In the present paper, the validated applied estimator remains the grouped Stage 1 posterior, while Stage 2 is retained as an exploratory extension. Algorithm 1 summarizes the end-to-end workflow.

IV. Results

Synthetic feasibility was first established in a controlled five-layer benchmark. In the clean synthetic setting, Stage 1 achieved approximately $R^2 = 0.976$ for load-related storage, $R^2 = 0.980$ for total water storage, $R^2 = 0.980$ for deformation reconstruction, and $R^2 = 0.981$ for groundwater. This synthetic block supports the inversion machinery under model-consistent conditions, while the real applied contribution of the paper lies in the grouped Bologna result; full synthetic and extended validation details are given in Supplementary Appendix D.

For the applied Emilia-Romagna case, conditioning diagnostics supported the grouped transition quantitatively. Relative to the earlier unconstrained five-layer real-data formulation, the grouped multisensor model reduced the stacked-operator condition number from approximately 2.16×10^6 to 6.99×10^1, reduced the information-matrix condition number from approximately 8.81×10^{10} to 4.88×10^3, restored practical rank from $4/5$ to $3/3$, and reduced the column-scale spread from approximately 1.26×10^6 to 6.70×10^1; see Supplementary Appendix C.

The current best real-data result is obtained from the grouped balanced multisensor Stage 1 model on the corrected Bologna MintPy/W3RA overlap, using InSAR deformation anomalies, GRACE regional anomalies, and refreshed SMAP surface-soil-moisture anomalies. Regional posterior skill reaches approximately $R^2 = 0.971$ for InSAR, $R^2 = 0.487$ for GRACE, and $R^2 = 0.618$ for SMAP, while the tiled posterior achieves an InSAR $R^2 = 0.905$. The grouped posterior also remains physically plausible in magnitude, with absolute maxima of approximately 52.1 mm for ShallowLoad, 88.1 mm for DeepLoad, and 48.1 mm for Groundwater. Figure 2 shows the grouped posterior maps and regional multisensor fits, while Table I summarizes the main quantitative metrics. SWOT was technically integrated but remained too sparse to materially improve the current grouped solution.

Independent validation was carried out by comparing the grouped posterior against Bologna groundwater wells in anomaly space using spatial interpolation, nearest-date alignment, and a discrete lag search over $-90, -60, -30, -14, 0, 14, 30, 60, 90$ days. Under this setup, 83 station series were evaluated. All 83 yielded positive best-match anomaly correlation; 62 reached correlation greater than or equal to 0.6; the median correlation was approximately 0.704; and the top-10 mean correlation was approximately 0.947. Depth-stratified validation gives median correlations of approximately 0.767 for shallow wells, 0.731 for intermediate wells, and 0.617 for deep wells. A trusted post-validation subset contains 21 stations across 6 hydrogeologic groups. Figure 3 summarizes the well-validation results, including representative time-series comparisons showing that the grouped groundwater posterior follows the observed well-head anomalies after lag optimization; the extended well-validation analysis is reported in Supplementary Appendix D.

TABLE I
MAIN QUANTITATIVE RESULTS FOR THE GROUPED BOLOGNA ESTIMATOR.

Metric	**Value**	**Metric**	**Value**
Regional InSAR posterior R^2	≈ 0.971	Tiled InSAR posterior R^2	≈ 0.905
Regional GRACE posterior R^2	≈ 0.487	Max $\lvert$ShallowLoad$\rvert$	52.1 mm
Regional SMAP posterior R^2	≈ 0.618	Max $\lvert$DeepLoad$\rvert$	88.1 mm
Evaluated wells	83	Max $\lvert$Groundwater$\rvert$	48.1 mm
Median well correlation	0.704	Trusted hydrogeologic groups	6
Top-10 mean corr.	0.947	Trusted stations	21
Wells with corr. ≥ 0.5	62		

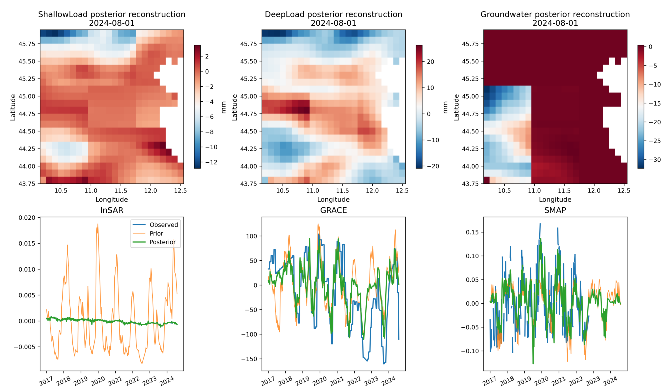

Fig. 2. Main grouped Stage 1 result for the Bologna overlap. Top: full-grid posterior reconstructions for ShallowLoad, DeepLoad, and Groundwater on the final overlap date. Bottom: observed, prior, and posterior regional time-series comparisons for InSAR, GRACE, and SMAP.

V. Conclusion

This work presents a novel physics-aware hybrid MCMC and Swin-transformer framework for grouped inversion of groundwater and shallow hydrologic storage from InSAR deformation. The central result is that hydrologic storage can be inferred directly from mixed high-resolution deformation data without prior explicit filtering of disturbing surface or subsurface processes. Synthetic feasibility demonstrates that the deformation-space inversion machinery is internally coherent, while conditioning analysis shows why grouped inversion is the stable real-data choice for Emilia-Romagna.

In the Bologna application, the grouped multisensor posterior remains physically plausible, improves the assimilated observation streams, and is independently supported by groundwater-well anomalies. The strongest external support is obtained for the grouped groundwater component, while the shallow hydrologic component is recovered at higher spatial resolution and supported through physically consistent posterior structure and multisensor agreement. These results support a practical route toward automated downstream satellite retrieval of regional and local hydrologic storage.

Reproducibility: A public GitHub repository for this study is available at `MSSJoud/insar-layered-groundwater-mcmc-swin`. The repository is intended to provide the containerized workflow, core scripts, configuration files, and figure-generation routines used in the paper, together with documentation for reproducing the grouped multisensor inversion and validation steps.

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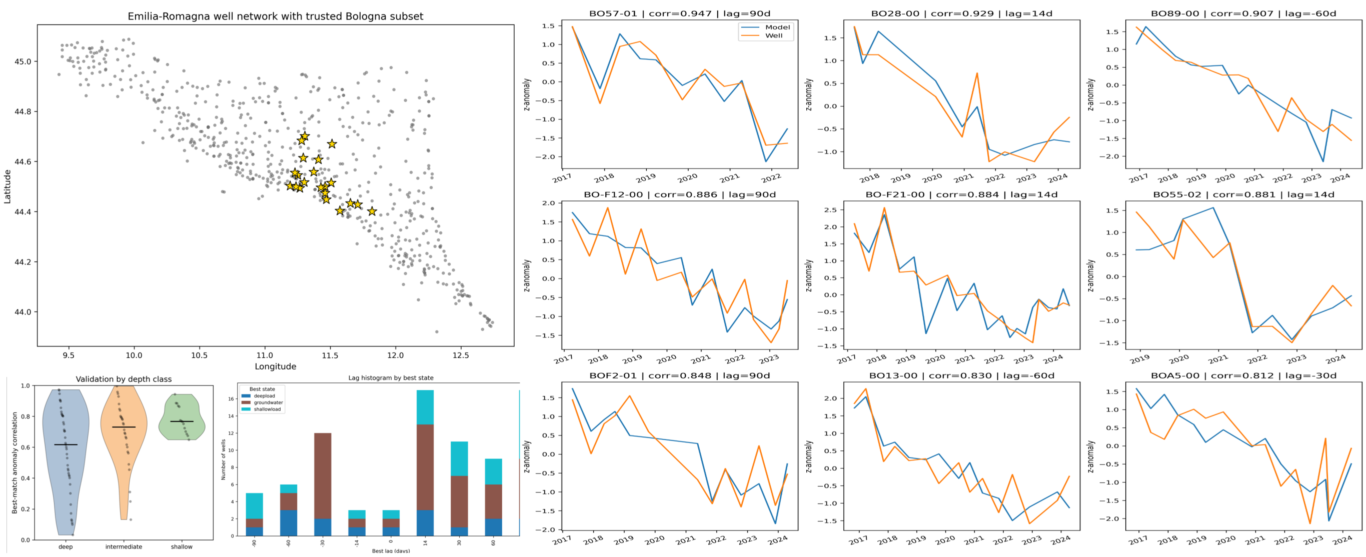

Fig. 3. Independent well validation of the grouped posterior. The map panel shows the Emilia-Romagna well network and the trusted Bologna subset, the summary panels show correlation and best-state behavior, and the lower panels give representative best-match time-series comparisons between grouped posterior anomalies and observed groundwater-well anomalies after lag optimization.

groundwater-monitoring data. The authors also acknowledge the developers and maintainers of the open-source tools used in this work, including MintPy and ISCE++.

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